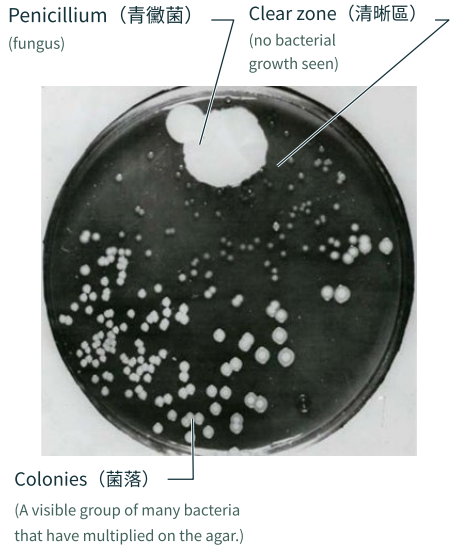
Name: _____ Class: _____ No.: _____

Antibiotic（抗生素） investigation

Record your initial observations. Let your questions guide the study.

The history of antibiotic（抗生素） discovery—an unexpected observation



In 1928, Fleming noticed contamination by Penicillium（青黴菌） on plates growing staphylococci, with reduced bacterial growth nearby. He investigated an antibacterial substance produced by this fungus. Later, Florey, Chain and their team advanced the therapeutic use of penicillin（青黴素）, an antibiotic（抗生素）.

Note: Fleming’s historical work involved staphylococci, not MRSA（耐甲氧西林金黃葡萄球菌）; MRSA（耐甲氧西林金黃葡萄球菌） was first formally reported in 1961.

Image source: https://www.cbsnews.com/news/almanac-the-discovery-of-penicillin（青黴素）/

Research context

A patient’s wound infection has not clearly improved after initial antibiotic（抗生素） treatment. Laboratory testing identifies methicillin-resistant Staphylococcus aureus (MRSA)（耐甲氧西林金黃葡萄球菌）. As a researcher, investigate how samples X, Y and Z affect visible growth of these bacteria in a virtual laboratory.

What is methicillin-resistant Staphylococcus aureus（耐甲氧西林金黃葡萄球菌）?

Resistant（具抗藥性的）: The bacteria resist a particular antibiotic（抗生素）, so it does not effectively kill them or inhibit（抑制） their growth.

Methicillin（甲氧西林）: The name of an antibiotic（抗生素） formerly used to treat some bacterial infections.

Methicillin-resistant Staphylococcus aureus (MRSA)（耐甲氧西林金黃葡萄球菌）: Staphylococcus aureus（金黃葡萄球菌） bacteria that resist methicillin（甲氧西林）. MRSA（耐甲氧西林金黃葡萄球菌） does not resist every antibiotic（抗生素）.

Main research question

Under the stated simulation conditions, how do X, Y and Z affect visible growth of these MRSA（耐甲氧西林金黃葡萄球菌） bacteria?

Your initial observations

Describe what you can see, without explaining the cause yet.

Make a prediction and consider the variables, assumptions and control before starting the experiment.

Complete your prediction

If discs containing antibiotic（抗生素） samples X, Y and Z are placed on an agar plate（瓊脂板）,（X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ none of the）discs will have a clear zone（清晰區） (a region with no visible bacterial growth) around them.

Among these,（X ／ Y ／ Z ／ Not applicable）will have the largest clear zone（清晰區） around its disc.

Not applicable (no clear zones（清晰區） predicted).

Circle one option in each set of brackets. Preserve your original responses.

Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)

Identify the variables

Choose the suitable factors in each category. Save your thinking now and view reference feedback after submission.

Tick all applicable boxes; more than one option may be selected.

Independent variable（The factor you change）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Dependent variable（The outcome you observe/measure）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Controlled variables（The factors kept the same）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Your original research plan

Each independent plate has X, Y, Z and the control. One more plate gives one more observation per sample. A higher number does not automatically mean a better plan.

There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)

（1 ／ 2 ／ 3 ／ 4 ／ 5 ／ 6 ／ 7 ／ 8 ／ 9 ／ 10 ／ 11 ／ 12 ／ 13 ／ 14 ／ 15 ／ 16 ／ 17 ／ 18 ／ 19 ／ 20）

Why would you plan it this way?

Preserve the original design in A before distributing B. Do not overwrite it.

What are the assumptions of this investigation?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Select all applicable assumptions; you may choose more than one.

- ☐ Bacteria are initially distributed evenly on each agar plate（瓊脂板）.
- ☐ All plates use the same simulated observation conditions.
- ☐ Equipment does not introduce contamination.
- ☐ If no colonies（菌落） form near a disc, this means all bacteria at that location have died.

The investigation control

If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?

(Hint: The control group provides a baseline for comparison. It helps check whether changes in the experiment are caused by the independent variable rather than other external factors. Therefore, the control group does not receive the treatment being tested.)

(Hint: how does the control differ from the experimental group?)

Draw and label your plates

Use the materials below as a reference. Label X, Y, Z and C (control), considering spacing between discs and from the plate edge. The beaker of alcohol and alcohol lamp are used to sterilise（滅菌／消毒） forceps, reducing contamination when picking up and placing discs.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.



Agar plate（瓊脂板） ×3 (contains nutrients needed for bacterial growth)



Marker ×1



Dropper ×1



Diluted（已稀釋的） bacterial suspension（細菌懸液） ×1



Sterile（無菌的） spreader（塗布器） ×1



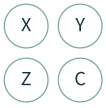
Forceps ×4



Beaker of alcohol ×1



Alcohol lamp ×1



Sample and control disc set ×3 (C = control)



Incubator（培養箱） ×1



Ruler ×1

Setup labels / written design

Add labels to your drawing, or give a complete written description of your plates as an alternative to drawing.

Preserve the original design in A before distributing B. Do not overwrite it.

Experiment and record

Follow the steps on the right to prepare three agar plates（瓊脂板）, incubate（培養） them together, then observe and measure.

This booklet provides fixed teaching-model images and raw diameter data for three independent plates. Complete the observations, means and conclusions yourself.

Your original research proposal has been saved separately. For classroom comparison, this activity will use 3 agar plates（瓊脂板）. You do not need to change your original design.

1 incubator（培養箱）: use the 30°C and 24-hour settings for the three plates together. X, Y, Z discs and a control: samples are carried on discs; the control has the same carrier alone.

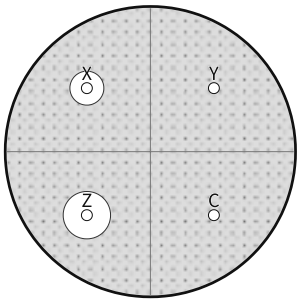


Plate 1

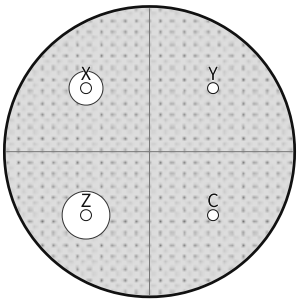


Plate 2

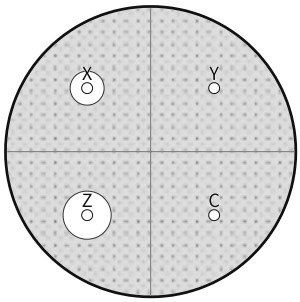


Plate 3

Measure and record the total clear-zone（清晰區） diameter (mm).

Plate	Sample	Raw diameter data (mm)	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.1	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	
2	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.3	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	
3	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.5	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	

Images and raw diameters are generated by the current VL4 result model. No colony（菌落） counts, drug concentrations or patient doses have been added. Raw data are not mean or conclusion answers.

VL ruler guidance (paper uses the pre-rendered data above): Drag the ruler to measure; ticks are 1 mm. Near a clear-zone（清晰區） diameter it holds that height while remaining free to move left and right. Pull farther up or down to release.

Procedure

1. Flip the agar plate（瓊脂板）, divide its bottom into four areas with a cross, and label X, Y, Z and C.
2. Flip the agar plate（瓊脂板） back; use the dropper to collect diluted（已稀釋的） bacterial suspension（細菌懸液） and release it at the centre.
3. Move the sterile（無菌的） spreader（塗布器） up and down and rotate the agar plate（瓊脂板） to distribute the bacteria evenly.
4. Select the forceps and use the alcohol beaker followed by the lamp to sterilise（滅菌／消毒） them. Pick up a disc and place it at its labelled centre. Repeat sterilisation（滅菌／消毒） before picking up every disc.
5. Cover the agar plate（瓊脂板） and place it upside down.
6. Repeat steps 1–5 for agar plates（瓊脂板） 2 and 3.
7. When all three plates are ready, place them in the incubator（培養箱）, select 30°C and 24 hours, and start incubation（培養）.
8. For plates 2 and 3, you may choose to complete steps 1–5 automatically.

Revisions, if any (keep the first response readable; this is not an additional question)

Analyse and conclude

Use your observations and measurements to answer the multiple-choice questions and draw a conclusion.

Mean total diameter calculation

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Mean for X (mm) _____ Mean for Y (mm) _____

Mean for Z (mm) _____ Mean for C (control) (mm) _____

1. What does the appearance of a clear zone（清晰區） indicate?

- ○ A. All bacteria in the zone have died.
- ○ B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
- ○ C. Bacteria in the zone resist every antibiotic（抗生素）.
- ○ D. The sample has eliminated viruses on the agar plate（瓊脂板）.

2. What does the diameter of a clear zone（清晰區） represent?

- ○ A. The exact number of bacteria remaining on the plate.
- ○ B. Its effectiveness in patients; the largest diameter identifies the best treatment.
- ○ C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
- ○ D. The antibiotic（抗生素） dose a patient should receive.

3. Draw a conclusion from your results:

Under these simulated conditions, （X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ No samples） can effectively inhibit（抑制） MRSA（耐甲氧西林金黃葡萄球菌） growth.

Comparing the mean total clear-zone（清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows:

（X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable）

Rank only samples with visible clear zones（清晰區）.

4. Why do experiments need to be repeated?

- ○ A. To ensure that every experiment produces exactly identical absolute values, with no variation whatsoever.
- ○ B. To prolong the experiment so scientists have more to describe when writing their reports.
- ○ C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
- ○ D. To make biological samples rapidly mutate and develop entirely new traits through repeated procedures.

5. If repeated results differ greatly, what should you do, and why?

- ○ A. Delete results that disagree with the original hypothesis so the results match.
- ○ B. Keep only the largest reading because it best represents inhibition（抑制）.
- ○ C. Calculate the mean immediately because averaging removes all procedural problems.
- ○ D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.

6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?

- ○ A. Ensure that all discs have clear zones（清晰區） of the same diameter.
- ○ B. Identify the best antibiotic（抗生素） for a patient.
- ○ C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.
- ○ D. Increase the amount of sample in the X, Y and Z discs to increase the accuracy（準確度） of the experimental results.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

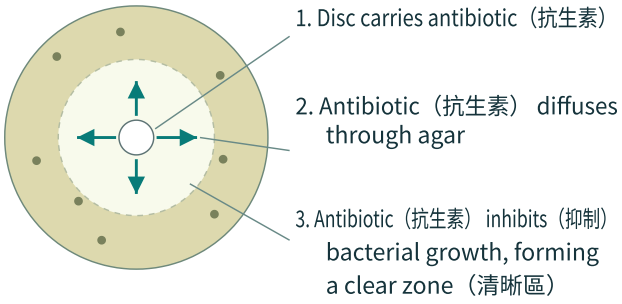
Revisions, if any (keep the first response readable; this is not an additional question)

Name: _____ Class: _____ No.: _____

Distribute this booklet only after the original inquiry answers have been fixed and preserved.

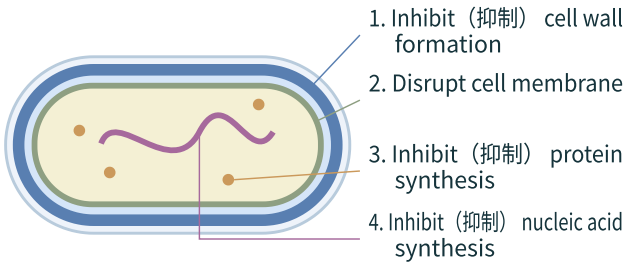
Learning points

A. Experimental principle: antibiotic（抗生素） diffusion and clear-zone（清晰區） formation



Schematic, not to scale; it does not prove complete killing.

B. Structures and possible action sites



The pale outer layer represents a capsule (not present in all bacteria); blue is the cell wall, green is the cell membrane, and the purple strand is DNA. These mechanisms are not proved by this disc experiment.

Uses of antibiotics（抗生素）

- Antibiotics（抗生素） mainly treat bacterial infections; they cannot treat viral colds or influenza.

How clear zones（清晰區） form

- Antibiotic（抗生素） diffuses outward from the disc, and its concentration generally decreases with distance. A clear zone（清晰區） forms only where the concentration remains sufficient to inhibit（抑制） this bacterium. Greater susceptibility（對抗生素的敏感性） (growth inhibited（抑制） at a lower antibiotic（抗生素） concentration) allows a larger clear zone（清晰區） to form.
- A clear zone（清晰區） indicates no visible bacterial growth at that location, demonstrating that the antibiotic（抗生素） sample can inhibit（抑制） bacterial growth; it does not prove that all bacteria in the zone have died.

Antibiotic resistance（抗生素抗性／抗藥性） and selection

- Why does the original antibiotic（抗生素） in the scenario lose effectiveness? Antibiotics（抗生素） make survival harder for susceptible（對抗生素敏感的） bacteria, while resistant（具抗藥性的） bacteria are more likely to survive, reproduce（繁殖） and pass on resistance traits（抗藥性特徵）. As resistant（具抗藥性的） bacteria increase in proportion and occupy the whole population, the original antibiotic（抗生素） becomes ineffective.
- Antibiotic（抗生素） overuse（過度使用） or misuse（不當使用） adds unnecessary selection pressure（選擇壓力）, promoting the survival and reproduction（繁殖） of resistant（具抗藥性的） bacteria and potentially making infections harder to treat.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

Distribute this booklet only after the original inquiry answers have been fixed and preserved.

Learning reflection

Your experimental results:

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.

Revisions, if any (keep the first response readable; this is not an additional question)

Teacher only: do not distribute to students

Distribution and research administration

1. Distribute A first. Preserve the original hypothesis, reasons, variables, assumptions, control, drawing and repeat plan. Keep a readable student copy before distributing B.
2. B uses three independent plates under shared teaching conditions of 30°C and 24 hours. Students interpret the images and raw data, record observations, calculate means and complete the six analysis items.
3. Collect or fix the original answers in A and B before distributing C. Reflection uses the student’ s own experimental results. C prints no reference conclusion or correctness marks.
4. D and the complete pack are for teachers only. Do not distribute the entire pack to students at once. Keep language, available time and prompts consistent（一致的） across groups.
5. This feature does not read or create student records or change answers, drawings, demonstrations, timing, events, cloud records or Excel. Researchers separately preserve and score handwritten paper data.
6. The supplied proposal, sections 4.1 and 4.3.2, specifies four weekly modules before formal teaching, with identical scenarios, concepts and prompts. Paper replaces interactive simulation with static diagrams and pre-rendered data tables. This booklet corresponds to VL4 antibiotics（抗生素）.
7. Paper uses fixed seeds; individual VL inquiries use independent seeds. The parameters and result model are the same, but individual raw values may differ. The researcher must confirm that this variation fits the study design.
8. Paper cannot measure virtual tool operation. Review cross-group SPS coverage, the 23-mark weighting, inter-rater agreement and separate formal SPS/concept pre/post tests. Trial-print B in greyscale and check image and text readability before use.
9. The proposal specifies 20 students per group, combining X1/X2: 11/9 in VL and 12/8 in paper. The separate 10-item SPSAI and 4-item CKT are administered in week 0 and after week 4, using the same pre/post tests. This worksheet does not replace those instruments or establish their validity.

These draft criteria have not been validated or calibrated across VLs. Speed, clicks, completion rate or a prediction matching the outcome are not measures of ability.

Images and raw diameters are generated by the current VL4 result model. No colony（菌落） counts, drug concentrations or patient doses have been added. Raw data are not mean or conclusion answers.

VL1 reference: v3 · No digestive-system questions are imported.

Teacher only: do not distribute to students

Reference answers

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.
1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

Raw diameter data (mm)

Plate	X	Y	Z	C (control)
1	18.7	0.0	26.1	0.0
2	18.7	0.0	26.3	0.0
3	18.7	0.0	26.5	0.0
Mean total diameter calculation	18.7	0.0	26.3	0.0

Visible zone outside the disc? Yes：X、Z；No：Y、C (control)。

Draw a conclusion from your results: X、Z。

Comparing the mean total clear-zone（清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows: Z ＞ X ＞ Not applicable ＞ Not applicable。

Score ranking against the student’ s own means; equal means allow either order.

Teacher only: do not distribute to students

Written responses: reference guidance

Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Which samples do you predict will have an area without visible bacterial growth around the disc?

A reasonable testable prediction may differ from the results. A reasoned prediction does not have to match the outcome.

Which sample do you predict will have the largest clear zone（清晰區） around its disc?

Give a reasoned prediction based on diffusion and bacterial growth. Choose not applicable if you predict no clear zones（清晰區）. The prediction need not match the result.

Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)

Explain a reason related to bacterial growth and observable evidence.

If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Setup labels / written design

Label each sample and control and allow enough space between discs and from the edge to avoid overlapping zones. A drawing or written setup is acceptable.

There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)

Include the first test. Three is not a required correct answer; more repeats do not guarantee reliability（可靠性）.

Why would you plan it this way?

Consider consistency（一致性）, variation, fair comparisons and practical feasibility.

Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.

Use readings for your chosen sample to support growth inhibition（抑制）. Explain how pre-existing resistance（抗藥性） affects survival and reproduction（繁殖）, how resistance traits（抗藥性特徵） are inherited（遺傳）, and how the resistant（具抗藥性的） proportion increases as antibiotic（抗生素） effectiveness declines. This is a hypothetical extension, not evidence that resistance（抗藥性） arose in this experiment.

These draft criteria have not been validated or calibrated across VLs. Speed, clicks, completion rate or a prediction matching the outcome are not measures of ability.

Teacher only: do not distribute to students

Scoring criteria (current VL4 draft)

These draft criteria have not been validated or calibrated across VLs. Speed, clicks, completion rate or a prediction matching the outcome are not measures of ability.

Item	Marks	Method	Full-credit criterion	Partial credit	Zero credit / limits
Initial observation	2	Teacher	Distinguish visible colonies（菌落）and the clear area near the fungus; describe observations without assuming a cause.	1: relevant description, but location or observation/inference distinction is incomplete.	No relevant observation, or only an unsupported causal claim.
Visible zones and diameters	2	Shared check	For each of 12 readings, use the correct Yes/No classification and a total diameter within ±1 mm; no visible zone requires exactly 0. Score matched readings × 2/12, rounded to two decimal places.	Proportional credit for matched readings.	No matching evidence; a 6 mm disc alone is not a clear zone（清晰區）.
Independent, dependent and controlled variables	4	Shared check	Select only sample for IV, zone for DV; controls are disc size, agar composition, strain, initial distribution and preset preparation.	IV: 1; DV: 1. Each correct control earns 2/5, rounded to two decimal places; an unrelated control choice gives 0 for the control item.	IV/DV must match the exact set.
Testable hypothesis and reason	1	Teacher	A testable prediction linked to bacterial growth and observable evidence.	0.5: a testable prediction with a weak or unclear reason.	No testable prediction or relevant reason. An unsupported prediction is not automatically wrong.
Original repeat-plan reason	1	Teacher	Explain repeats through consistency（一致性）, variation, fair comparison and feasibility.	0.5: mentions reliability（可靠性）but lacks independent comparison or a rationale for the number.	Only a number or no relevant reason; choosing three is not a required answer.
Control design and reason	1	Teacher	Use a matching carrier-only disc without antibiotic（抗生素）; hold other conditions constant and explain the comparison.	0.5: suitable C, but the reason or shared conditions are incomplete.	No useful comparison or confounded treatment.

Teacher only: do not distribute to students

Scoring criteria (current VL4 draft)

These draft criteria have not been validated or calibrated across VLs. Speed, clicks, completion rate or a prediction matching the outcome are not measures of ability.

Item	Marks	Method	Full-credit criterion	Partial credit	Zero credit / limits
Assumptions	1	Shared check	Select only even initial distribution, no added contamination and shared observation conditions.	No partial credit.	Any missing or additional choice, including assuming all bacteria died.
Plate-design quality	2	Teacher	Label X, Y, Z and C clearly; allow space between discs and from the edge. A drawing or complete written design is acceptable.	1: basically feasible but missing one important arrangement.	No assessable design; artistic quality is not a criterion.
Means	1	Shared check	Use the student’s own three latest readings; ±0.1 mm tolerance; 0.25 per sample.	Proportional credit across four samples.	Do not penalise the same measurement error again through arithmetic.
Reflection: evidence and resistance（抗藥性） selection	2	Teacher	Cite sample readings to support inhibition（抑制）; link prior resistance（抗藥性） differences, survival and reproduction（繁殖）, inherited（遺傳） traits and changing population proportions to loss of effectiveness.	1: a reasonable experimental judgment or selection explanation, but missing readings or causal links.	No relevant evidence/concept, or bacteria purposefully developing resistance（抗藥性） to survive.
Analysis 1: What does the appearance of a clear zone（清晰區） indicate?	1	Shared check	Correct option: B.	No partial credit.	Incorrect or missing response: 0.
Analysis 2: What does the diameter of a clear zone（清晰區） represent?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.

Teacher only: do not distribute to students

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Analysis 4: Why do experiments need to be repeated?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.
Analysis 5: If repeated results differ greatly, what should you do, and why?	1	Shared check	Correct option: D.	No partial credit.	Incorrect or missing response: 0.
Analysis 6: When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.
Selected conclusion	1	Shared check	X and Z: 0.5. Rank effective samples by the student’ s own means: 0.5. Unused ranks are Not applicable.	0.5 if either component is correct.	Either order is accepted for equal means; do not penalise measurement error again.
Current total / pending marks	23	Formula	Total 23. All six manual fields must be scored and reflection submitted before a total is reported.	Blank manual marks mean pending; 0 means marked zero.	Do not compare directly with the legacy 32-mark rubric. Draft calibration is required.